

Axioms for the real numbers: arithmetic and order

Addition

- (A1) if $x, y \in \mathbb{R}$ then $x + y \in \mathbb{R}$ (closure under addition)
- (A2) if $x, y, z \in \mathbb{R}$ then $(x + y) + z = x + (y + z)$ (addition is associative)
- (A3) if $x, y \in \mathbb{R}$ then $x + y = y + x$ (addition is commutative)
- (A4) there exists $0 \in \mathbb{R}$ such that for any $x \in \mathbb{R}$, $x + 0 = 0 + x = x$ (existence of an additive identity)
- (A5) if $x \in \mathbb{R}$ there exists $-x \in \mathbb{R}$ such that $x + (-x) = (-x) + x = 0$ (existence of an additive inverse)

Multiplication

- (M1) if $x, y \in \mathbb{R}$ then $xy \in \mathbb{R}$ (closure under multiplication)
- (M2) if $x, y, z \in \mathbb{R}$ then $(xy)z = x(yz)$ (multiplication is associative)
- (M3) if $x, y \in \mathbb{R}$ then $xy = yx$ (multiplication is commutative)
- (M4) there exists $1 \in \mathbb{R}$, such that for any $x \in \mathbb{R}$, $1.x = x.1 = x$ (existence of multiplicative identity)
- (M5) if $x \in \mathbb{R}$ with $x \neq 0$, there exists $x^{-1} \in \mathbb{R}$ such that $x.x^{-1} = x^{-1}.x = 1$ (existence of multiplicative inverses)

Interaction between addition and multiplication

- (D) if $x, y, z \in \mathbb{R}$, $x(y + z) = xy + xz$ (distributive property)

Order properties. We use $y > x$ as an equivalent for $x < y$, and $x \leq y$ means that either $x < y$ or $x = y$.

- (O1) if $x, y \in \mathbb{R}$ exactly one of the statements $x < y$, $x = y$, $x > y$ is true (trichotomy)
- (O2) if $x, y, z \in \mathbb{R}$ then if $x < y$ and $y < z$, $x < z$ (transitivity)

Interaction between order and arithmetic

- (O3) if $x, y, z \in \mathbb{R}$ and $x < y$, then $x + z < y + z$
- (O4) if $x, y, z \in \mathbb{R}$, $x < y$, and $z > 0$, then $xz < yz$